

7. (a) Carys and Pavel prepared crystals of zinc sulfate by reacting zinc carbonate with an acid.
- (i) Give the name of the acid they used. [1]
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- (ii) In the first stage of their preparation, they added excess zinc carbonate to the acid.
- I. Give the observation that **immediately** shows a reaction is taking place. [1]
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- II. State why they added **excess** zinc carbonate. [1]
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- (iii) Describe the remaining two stages they carried out to obtain a **pure** sample of zinc sulfate crystals. [2]
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- (iv) Give the chemical formula of zinc sulfate. [1]
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- (b) In another experiment, Carys and Pavel investigated the temperature rise when dilute hydrochloric acid neutralises sodium hydroxide solution.



The acid was added 5 cm^3 at a time to 25 cm^3 of sodium hydroxide solution. They recorded the highest temperature reached after each addition using a digital thermometer.

They obtained the following results. The result for 15 cm^3 of acid is missing.

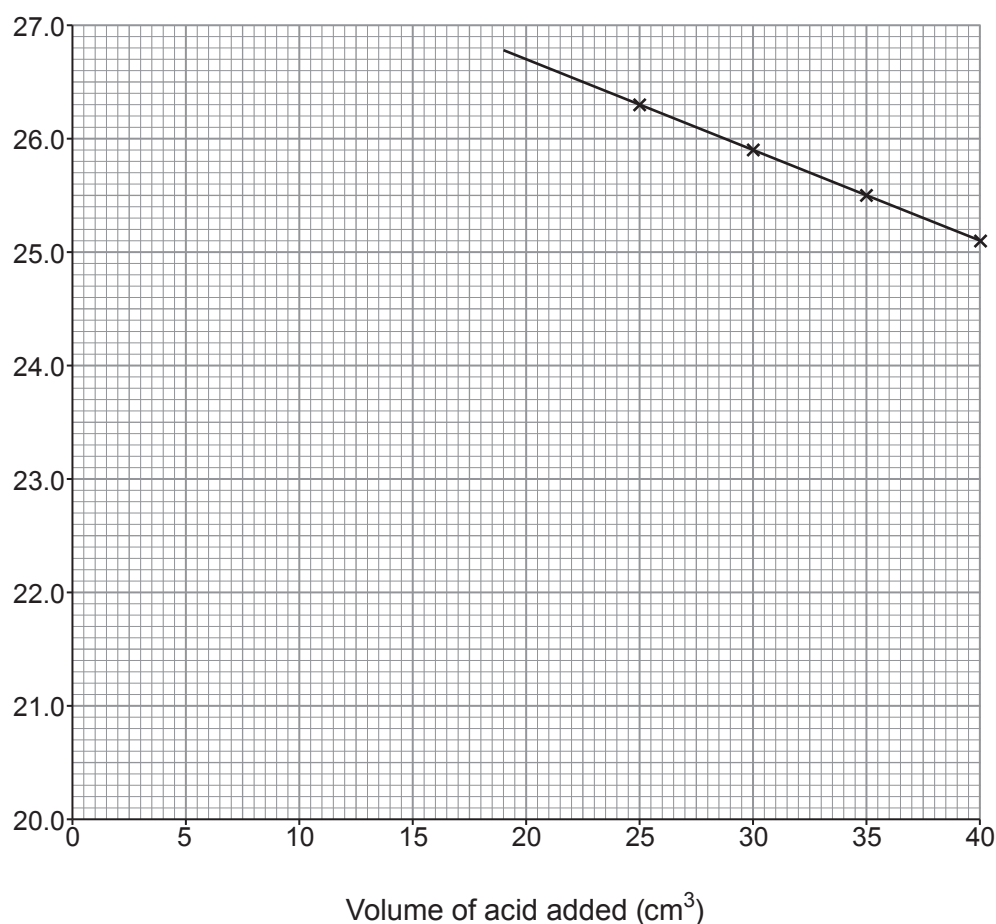
Volume of acid added (cm^3)	Temperature ($^{\circ}\text{C}$)
0	20.4
5	21.9
10	23.4
15	
20	26.4
25	26.3
30	25.9
35	25.5
40	25.1



- (i) The last four results have been plotted on the grid below and a straight line drawn through the points.

Plot the remaining four points and draw a **straight line** through them so that it intersects the line already drawn. [3]

Temperature ($^{\circ}\text{C}$)



- (ii) I. Use your graph to give the temperature that would have been recorded when 15 cm^3 of acid was added. [1]

..... $^{\circ}\text{C}$

- II. Carys and Pavel concluded that the volume of acid needed to just neutralise all the sodium hydroxide solution was somewhere between 20 cm^3 and 25 cm^3 .

Use the graph to suggest the exact volume of acid needed. [1]

..... cm^3



- (iii) The temperatures recorded are slightly **lower** than expected.

Tick (✓) to show which **two** improvements to the method would enable Carys and Pavel to obtain results closer to the expected values. [2]

use a beaker instead of a flask

☐

repeat the method

☐

add the acid in smaller intervals

☐

wrap cotton wool around the flask

☐

use a larger flask

☐

place a lid on the flask

☐

- (c) In a different experiment, it was found that the maximum temperature was reached when 40 cm^3 of hydrochloric acid was added to 20 cm^3 of sodium hydroxide solution.

State what this means in terms of the relative concentrations of the solutions used. [2]

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END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		





THE PERIODIC TABLE

Group

1 2 3 4 5 6 7 0

1 H Hydrogen 1																	
28 28																	
4 He Helium 2																	
20 Ne Neon 10																	
40 Ar Argon 18																	
84 Kr Krypton 36																	
131 Xe Xenon 54																	
222 Rn Radon 86																	
8 O Oxygen 8																	
16 S Sulfur 16																	
32 Se Selenium 34																	
79 Br Bromine 35																	
127 I Iodine 53																	
210 Po Polonium 84																	
210 At Astatine 85																	
12 C Carbon 6																	
14 N Nitrogen 7																	
31 P Phosphorus 15																	
75 As Arsenic 33																	
122 Sb Antimony 51																	
207 Pb Lead 82																	
209 Bi Bismuth 83																	
11 B Boron 5																	
27 Al Aluminium 13																	
73 Ge Germanium 32																	
70 Ga Gallium 31																	
65 Zn Zinc 30																	
63.5 Cu Copper 29																	
108 Ag Silver 47																	
197 Au Gold 79																	
195 Pt Platinum 78																	
192 Ir Iridium 77																	
190 Os Osmium 76																	
186 Re Rhenium 75																	
184 W Tungsten 74																	
181 Ta Tantalum 73																	
179 Hf Hafnium 72																	
139 La Lanthanum 57																	
227 Ac Actinium 89																	
88 Ra Radium 88																	
87 Fr Francium 87																	
226 Ra Radium 88																	
137 Ba Barium 56																	
86 Sr Strontium 38																	
88 Y Yttrium 39																	
91 Zr Zirconium 40																	
93 Nb Niobium 41																	
96 Mo Molybdenum 42																	
99 Tc Technetium 43																	
101 Ru Ruthenium 44																	
103 Rh Rhodium 45																	
106 Pd Palladium 46																	
112 Cd Cadmium 48																	
115 In Indium 49																	
127 Te Tellurium 52																	
128 Se Selenium 34																	
79 Br Bromine 35																	
80 Kr Krypton 36																	
35.5 Cl Chlorine 17																	
19 F Fluorine 9																	
20 Ne Neon 10																	

Key

A_r

Symbol

Name

Z

relative atomic mass

atomic number